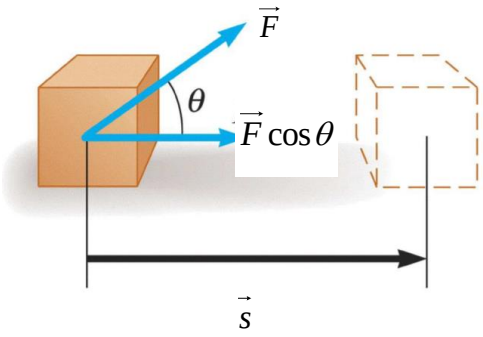
Scalar Product (Dot Product)

Vectors are widely used in physics, especially in mechanics (力學). For example, the work W done by a constant force $\vec{F}$ in moving an object through a displacement $\vec{s}$ is given by

$$W = |\vec{F}||\vec{s}|\cos\theta$$

(note that W is a scalar)

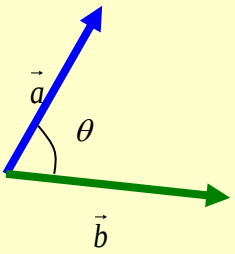


Since the above operation is applied frequently, in order to have a better represent, **scalar product (純量積) / dot product (點積)** is introduced, i.e.

Scalar Product (Dot Product)

The **scalar product** (also refer as **dot product**) of two vectors $\vec{a}$ and $\vec{b}$, denoted as $\vec{a}\cdot\vec{b}$, is defined as:

$$\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta$$



where θ is the angle between $\vec{a}$ and $\vec{b}$, and $0° \le \theta \le 180°$.

With scalar product, we can express the work W done by a constant force $\vec{F}$ in moving an object through a displacement $\vec{s}$ by $W = \vec{F}\cdot\vec{s}$.

Moreover, we can find the angle between two vectors $\vec{a}$ and $\vec{b}$ by $\cos\theta = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$.

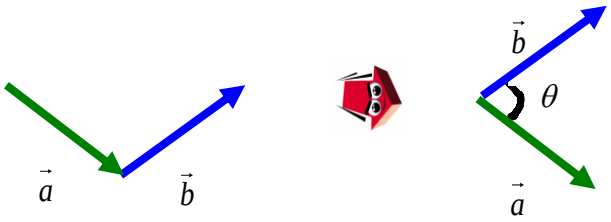
As a result, scalar product plays an important role in vector operation.

Not for sale

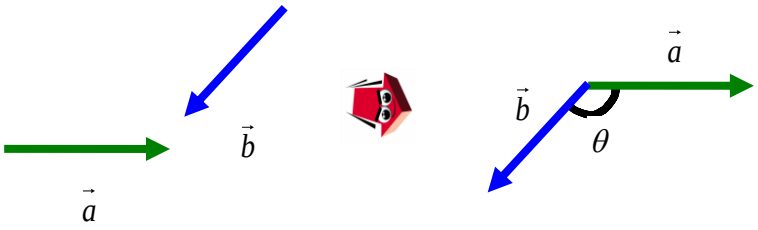
Definition of Scalar Product

For any two free vectors, we can locate the angle between them as follow:

e.g.1



e.g.2



As $\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta$, where θ is the angle between $\vec{a}$ and $\vec{b}$ $(0° \le \theta \le 180°)$

$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}, \quad \text{where } 0° \le \theta \le 180°.$$

Note that:

- $\vec{a}\cdot\vec{b}$ is positive when θ is acute, i.e. $0° \le \theta < 90°$.
- $\vec{a}\cdot\vec{b}$ is negative when θ is obtuse, i.e. $90° < \theta \le 180°$.
- $\vec{a}\cdot\vec{b}$ is zero when θ is right angle, i.e. $\theta = 90°$.

Furthermore, for any **vectors** $\vec{a}$, $\vec{b}$, $\vec{c}$ and **constants** k_1, k_2, we have the following properties

(i) $\vec{a}\cdot\vec{a} = |\vec{a}|^2$

(ii) $\vec{a}\cdot\vec{b} = \vec{b}\cdot\vec{a}$

(iii) $k_1(\vec{a}\cdot\vec{b}) = (k_1\vec{a})\cdot\vec{b} = \vec{a}\cdot(k_1\vec{b})$

(iii) $(k_1\vec{a})\cdot(k_2\vec{b}) = k_1k_2\,\vec{a}\cdot\vec{b}$

(iv) $\vec{a}\cdot(\vec{b}+\vec{c}) = \vec{a}\cdot\vec{b} + \vec{a}\cdot\vec{c}$

Not for sale

Definition of Scalar Product

[Add. Math 2009 Q.7]

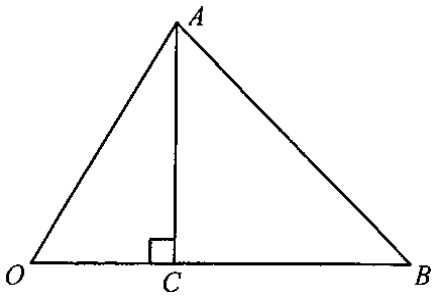


Figure 1

In Figure 1, AC is the altitude of ΔOAB. Let **a** , **b** and **c** be $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ respectively. It is given that |**a**| = 6, |**b**| = 8 and **a** · **b** = 24. Find

(a) $\angle AOB$,

(b) |**c**|. ***(4 marks)***

Applications of Dot Product (I) : Magnitude of vectors

For any vector $\vec{a}$, we have $\vec{a}\cdot\vec{a} = |\vec{a}||\vec{a}|\cos 0° = |\vec{a}|^2$.

Therefore we can find the **magnitude** of any vector by using dot product as follows:

$$|\vec{a}| = \sqrt{\vec{a}\cdot\vec{a}}$$

And it is useful to find the **magnitude** of any composite vector, such as $\vec{u} + \vec{v}$:

$$\begin{aligned}|\vec{u}+\vec{v}| &= \sqrt{(\vec{u}+\vec{v})\cdot(\vec{u}+\vec{v})}\\ &= \sqrt{\vec{u}\cdot\vec{u}+2\vec{u}\cdot\vec{v}+\vec{v}\cdot\vec{v}}\\ &= \sqrt{|\vec{u}|^2+2\vec{u}\cdot\vec{v}+|\vec{v}|^2}\end{aligned}$$

e.g. If $|\overrightarrow{AB}| = 3$, $|\overrightarrow{AC}| = 1$ and $\angle CAB = 60°$.

Find $|\overrightarrow{AB} + 2\overrightarrow{AC}|$.

$$\overrightarrow{AB}\cdot\overrightarrow{AC} = (3)(1)\cos 60° = \frac{3}{2}$$

$$\begin{aligned}&|\overrightarrow{AB}+2\overrightarrow{AC}|\\ &= \sqrt{(\overrightarrow{AB}+2\overrightarrow{AC})\cdot(\overrightarrow{AB}+2\overrightarrow{AC})}\\ &= \sqrt{\overrightarrow{AB}\cdot\overrightarrow{AB}+4\overrightarrow{AB}\cdot\overrightarrow{AC}+4\overrightarrow{AC}\cdot\overrightarrow{AC}}\\ &= \sqrt{|\overrightarrow{AB}|^2+4(\overrightarrow{AB}\cdot\overrightarrow{AC})+4|\overrightarrow{AC}|^2}\\ &= \sqrt{(3)^2+4(\frac{3}{2})+4(1)^2}\\ &= \sqrt{19}\end{aligned}$$



Always use the dot product to find the magnitude of a composite vector.

Not for sale

Applications of Dot Product (I) : Magnitude of vectors

[Add. Math 2006 Q.7]

Let **a** and **b** be vectors such that $|\mathbf{a}| = \sqrt{3}$, $|\mathbf{b}| = 2$ and the angle between them is 150°.

(a) Find $\mathbf{a} \cdot \mathbf{b}$.

(b) Find $|\mathbf{a} + 2\mathbf{b}|$. ***(5 marks)***

[M2 2017 Q.3]

P is a point lying on *AB* such that *AP* : *PB* = 3 : 2. Let $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$, where *O* is the origin

(a) Express $\overrightarrow{OP}$ in terms of **a** and **b**.

(b) It is given that $|\mathbf{a}| = 45$, $|\mathbf{b}| = 20$ and $\cos \angle AOB = \frac{1}{4}$. Find

(i) $\mathbf{a} \cdot \mathbf{b}$.

(ii) $\left|\overrightarrow{OC}\right|$. ***(5 marks)***

Not for sale

Applications of Dot Product (II) : Orthogonal vectors

Two vectors are said to be **orthogonal (正交)** if they are perpendicular to each other.

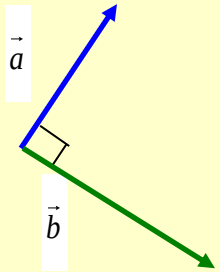
i.e. if the angle between non-zero vectors $\vec{a}$ and $\vec{b}$ are 90°, then $\vec{a}$ and $\vec{b}$ are orthogonal.

In this situation, $\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos 90° = 0$.

As a result, dot product is usually applied to test whether two vectors are orthogonal:

For any non-zero vectors $\vec{a}$ and $\vec{b}$,

$$\vec{a} \perp \vec{b} \quad \Leftrightarrow \quad \vec{a}\cdot\vec{b} = 0$$



e.g. If $|\vec{a}| = 4$, $|\vec{b}| = 3$ and the angle between $\vec{a}$ and $\vec{b}$ is 60°.

Show that $(3\vec{a} + \vec{b}) \perp (\vec{a} - 2\vec{b})$.

$\vec{a}\cdot\vec{b} = (4)(3)\cos 60° = 6$

$(3\vec{a}+\vec{b})\cdot(\vec{a}-2\vec{b})$

$= 3\vec{a}\cdot\vec{a} - (3\vec{a})\cdot(2\vec{b}) + \vec{b}\cdot\vec{a} - \vec{b}\cdot(2\vec{b})$

$= 3|\vec{a}|^2 - 5(\vec{a}\cdot\vec{b}) - 2|\vec{b}|^2$

$= 3(4)^2 - 5(6) - 2(3)^2$

$= 0$

$\therefore\ (3\vec{a} + \vec{b}) \perp (\vec{a} - 2\vec{b})$

Use dot product to expand the expression first.

$\vec{u}\cdot\vec{v} = 0 \Rightarrow \vec{u} \perp \vec{v}$

Not for sale

Applications of Dot Product (II) : Orthogonal vectors

[Add. Math 2001 Q.8]

Let **a**, **b** be two vectors such that $|\mathbf{a}| = 4$, $|\mathbf{b}| = 3$ and the angle between **a** and **b** is 60°.

(a) Find $\mathbf{a} \cdot \mathbf{b}$.

(b) Find the value of k if the vectors $(\mathbf{a} + k\mathbf{b})$ and $(\mathbf{a} - 2\mathbf{b})$ are perpendicular to each other.

(6 marks)

[Add. Math 1999 Paper I Q.7]

Let **a, b** be vectors such that $\mathbf{a} = 3\mathbf{i} + 4\mathbf{j}$ and $|\mathbf{b}| = 4$. The angle between **a** and **b** is 60°.

(a) Find $|\mathbf{a}|$.

(b) Find $\mathbf{a} \cdot \mathbf{b}$.

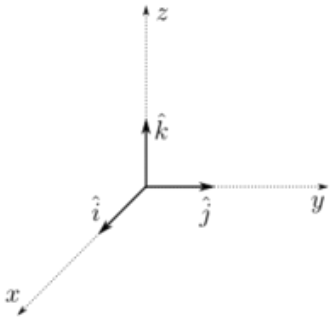
(c) lf the vector $(m\mathbf{a} + \mathbf{b})$ is perpendicular to **b,** find the value of m.

(6 marks)

Dot Product for Base Unit Vectors

In the coordinates system, there are three **base unit vectors**:

- $\vec{i}$ represents the unit vector in the **+ve direction of *x*-axis**
- $\vec{j}$ represents the unit vector in the **+ve direction of *y*-axis**
- $\vec{k}$ represents the unit vector in the **+ve direction of *z*-axis**



And there are some basic properties of base unit vectors:

(i) $|\vec{i}| = |\vec{j}| = |\vec{k}| = 1$

(ii) $\vec{i}\cdot\vec{i} = \vec{j}\cdot\vec{j} = \vec{k}\cdot\vec{k} = 1$

(iii) $\vec{i}\cdot\vec{j} = \vec{j}\cdot\vec{i} = 0$; $\vec{j}\cdot\vec{k} = \vec{k}\cdot\vec{j} = 0$; $\vec{i}\cdot\vec{k} = \vec{k}\cdot\vec{i} = 0$

Therefore there are two important results about dot product:

Let $\vec{u} = x_1\vec{i} + y_1\vec{j} + z_1\vec{k}$ and $\vec{v} = x_2\vec{i} + y_2\vec{j} + z_2\vec{k}$,

θ be the angle between $\vec{u}$ and $\vec{v}$ (where $0° \le \theta \le 180°$)

Then

- $\vec{u}\cdot\vec{v} = x_1x_2 + y_1y_2 + z_1z_2$
- $\cos\theta = \dfrac{\vec{u}\cdot\vec{v}}{|\vec{u}||\vec{v}|} = \dfrac{x_1x_2 + y_1y_2 + z_1z_2}{\sqrt{(x_1)^2 + (y_1)^2 + (z_1)^2}\sqrt{(x_2)^2 + (y_2)^2 + (z_2)^2}}$

e.g. If $\vec{u} = -\vec{i} - 3\vec{j} + 7\vec{k}$ and $\vec{v} = 4\vec{i} + \vec{j} + 5\vec{k}$,

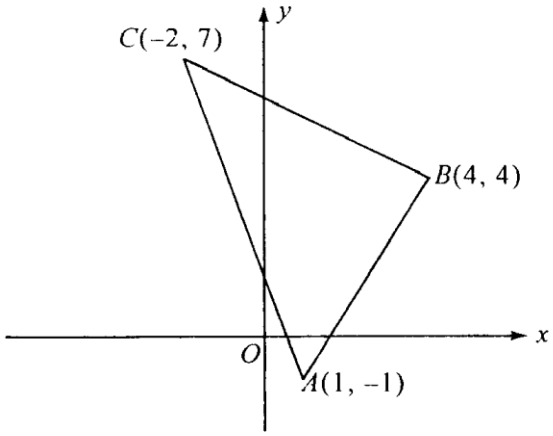
then $\cos\theta = \dfrac{(-1)(4) + (-3)(1) + (7)(5)}{\sqrt{(-1)^2 + (-3)^2 + (7)^2}\sqrt{(4)^2 + (1)^2 + (5)^2}}$

$\theta = 55.8°$ (corr. To 3 sig. fig.)

Not for sale

Dot Product for Unit Base Vectors

[Add. Math 1998 Paper I Q.5]



The points *A*, *B* and *C* whose position vectors are **i** – **j**, 4**i** + 4**j** and –2**i** + 7**j** respectively are shown in above figure.

(a) Find the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

(b) By considering $\overrightarrow{AB} \cdot \overrightarrow{AC}$, find $\angle BAC$ to the nearest degree. ***(6 marks)***

Dot Product for Unit Base Vectors

[M2 2017 Q.10(a)(b)]

ABC is a triangle. *D* is the mid-point of *AC*. *E* is a point on *BC*, where *BE* : *EC* = 1 : *r*. *AB* produced and *DE* produced meet at the point *F*. It is given that *DE* : *EF* = 1 :10.
Let $\overrightarrow{OA}$=2**i** + 3**j** – 2**k**, $\overrightarrow{OB}$ = 4**i** + 4**j** – **k**, $\overrightarrow{OC}$ = 8**i** – 3**j** – 2**k**, where *O* is the origin.

(a) By expressing $\overrightarrow{AE}$ and $\overrightarrow{AF}$ in terms of *r*, find *r*. ***(4 marks)***

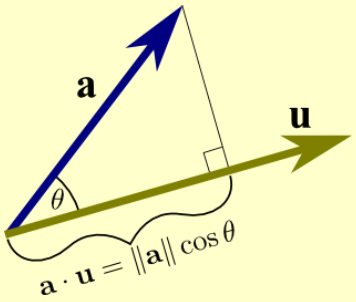
(b) (i) Find $\overrightarrow{AD} \cdot \overrightarrow{DE}$.

(ii) Are *B*, *D*, *C* and *F* concyclic? Explain your answer. ***(5 marks)***

Not for sale

Applications of Dot Product (III) : Projections

1. Scalar projection of $\vec{a}$ on $\vec{u}$



$$= |\vec{a}| \cos\theta$$

$$= \frac{|\vec{a}||\vec{u}| \cos\theta}{|\vec{u}|}$$

$$= \frac{\vec{a} \cdot \vec{u}}{|\vec{u}|}$$

(Note: $\frac{\vec{a} \cdot \vec{u}}{|\vec{u}|}$ is positive when $0° < \theta < 90°$; negative when $90° < \theta < 180°$)

2. Vector projection of $\vec{a}$ on $\vec{u}$

$$= \left(\frac{\vec{a} \cdot \vec{u}}{|\vec{u}|} \right) \hat{u}$$ where $\hat{u}$ is the unit vector in the direction of $\vec{u}$

$$= \left(\frac{\vec{a} \cdot \vec{u}}{|\vec{u}|} \right) \frac{\vec{u}}{|\vec{u}|}$$

$$= \left(\frac{\vec{a} \cdot \vec{u}}{|\vec{u}|^2} \right) \vec{u}$$

e.g. If $\vec{a} = 4\vec{i} - 2\vec{j} - 3\vec{k}$ and $\vec{u} = 4\vec{i} + 3\vec{j} - \vec{k}$,

then **vector projection of $\vec{a}$ on $\vec{u}$**

$$= \left(\frac{(4)(4) + (-2)(3) + (-3)(-1)}{\sqrt{(4)^2 + (3)^2 + (-1)^2}^2} \right) \vec{u}$$

$$= \frac{1}{2} \vec{u}$$

$$= 2\vec{i} + \frac{3}{2}\vec{j} - \frac{1}{2}\vec{k}$$

Not for sale

Applications of Dot Product (III) : Projections

[Add. Math 2007 Q.8 Modified]

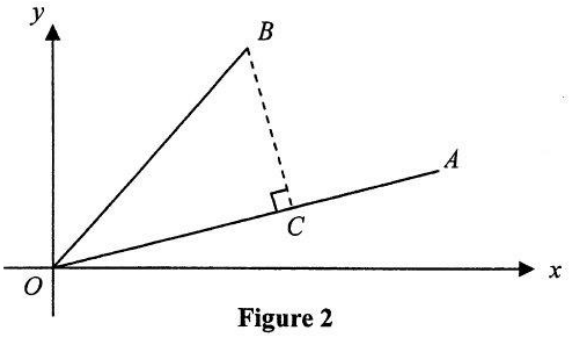


Figure 2

In Figure 2, *OCA* is a straight line and $BC \perp OA$. It is given that $\overrightarrow{OA} = 6\mathbf{i} + 3\mathbf{j}$ and $\overrightarrow{OB} = 2\mathbf{i} + 6\mathbf{j}$. Let $\overrightarrow{OC} = k\overrightarrow{OA}$.

Find the value of *k*. ***(6 marks)***

Applications of Dot Product (III) : Projections

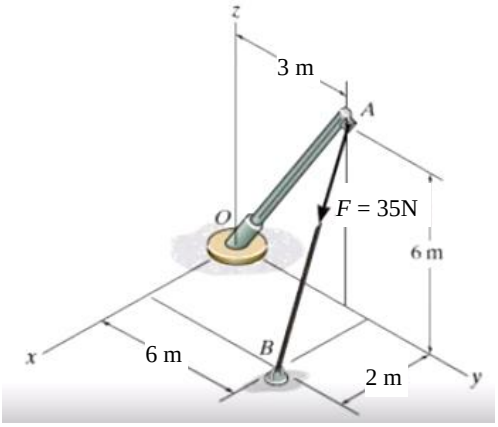
[Real-life example]

In the figure, a pipe *AO* is fixed on the ground at *O*. A force of 35N in the direction *AB* is acted on the pipe. Let the coordinates of *O* be (0, 0, 0).

(a) Find $\overrightarrow{AB}$. Hence find the angle between the pipe *AO* and force $\vec{F}$.

(b) If 1 unit of magnitude is equivalent to 1 N, find the force $\vec{F}$.

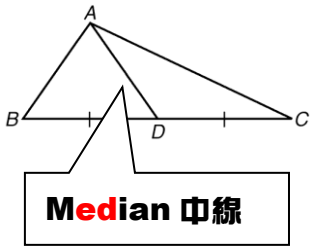
(c) Find the vector projection of the force $\vec{F}$ onto the pipe *AO*.



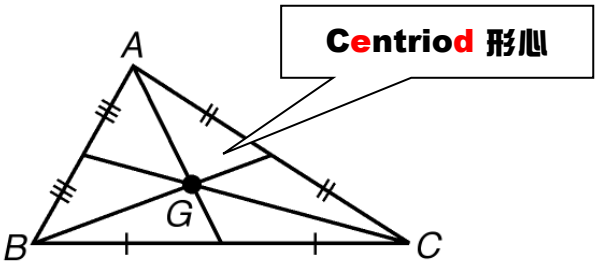
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Bonus : Centers of Triangles

1.

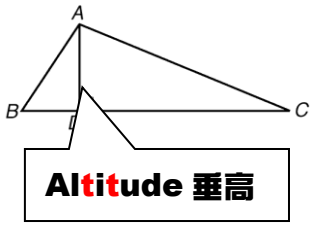


Median 中線

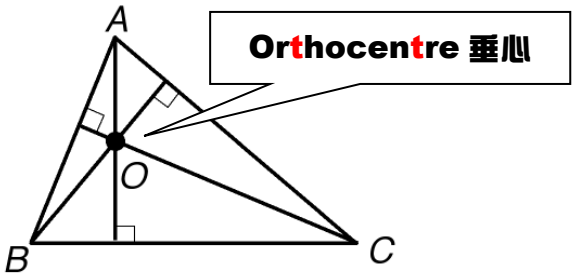


Centriod 形心

2.

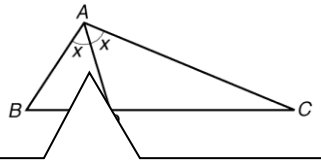


Altitude 垂高

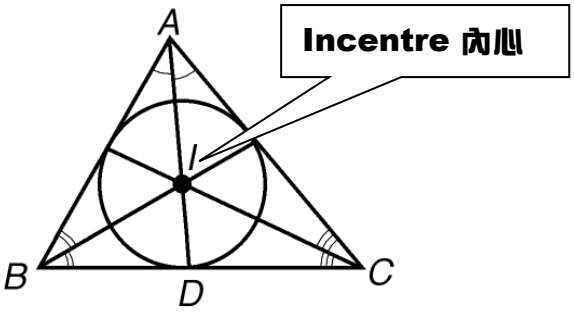


Orthocentre 垂心

3.

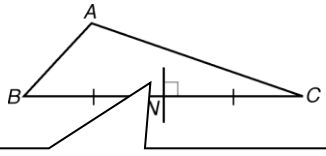


Angle bisector 角平分線

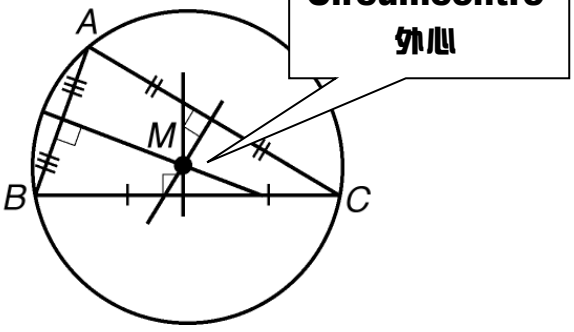


Incentre 內心

4.



Perpendicular bisector
垂直平分線



Circumcentre
外心

Not for sale

Bonus : Centers of Triangles

[M2 2012 Q.12]

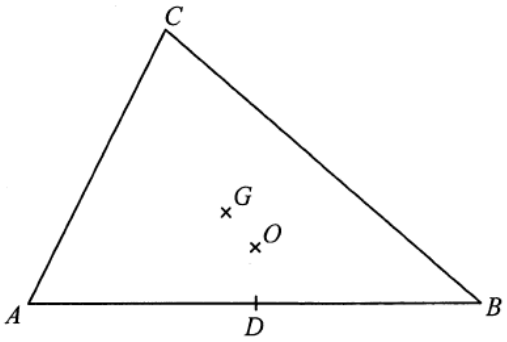


Figure 6

Figure 6 shows an acute angled scalene triangle *ABC*, where *D* is the mid-point of *AB*, *G* is the centroid and *O* is the circumcentre. Let $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$ and $\overrightarrow{OC} = \mathbf{c}$.

(a) Express $\overrightarrow{AG}$ in terms of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$. ***(3 marks)***

(b) It is given that *E* is a point on *AB* such that *CE* is an altitude. Extend *OG* to meet *CE* at *F*.

(i) Prove that $\Delta DOG \sim \Delta CFG$.

Hence find $FG : GO$.

(ii) Show that $\overrightarrow{AF} = \mathbf{b} + \mathbf{c}$.

Hence prove that *F* is the orthocenter of ΔABC. ***(9 marks)***

Not for sale



Bonus : Centers of Triangles

Not for sale

HKDSE Questions (level 2)

1. ***[M2 2013 Q.10]***

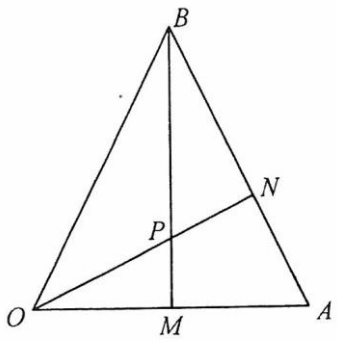


Figure 2

Let $\overrightarrow{OA} = 2\mathbf{i}$ and $\overrightarrow{OB} = \mathbf{i} + 2\mathbf{j}$. M is the mid-point of OA and N lies on AB such that $BN : NA = k : 1$. BM intersects ON at P (see Figure 2).

(a) Express $\overrightarrow{ON}$ in terms of k .

(b) If A , N , P and M are concyclic, find the value of k . ***(5 marks)***

HKCEE Questions (level 3)

2. ***[Add .Math 2002 Q.13]***

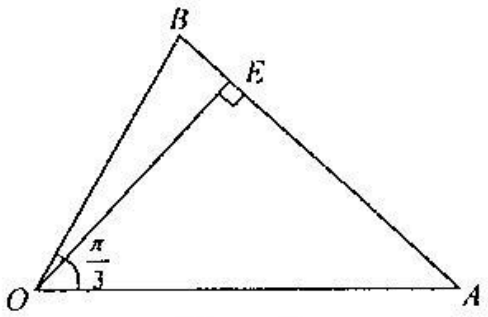


Figure 4

In Figure 4, *OAB* is a triangle. Point *E* is the foot of perpendicular from *O* to *AB*. Let $\overrightarrow{OA}$ = **a** and $\overrightarrow{OB}$ = **b**. It is given that *OA* = 3, *OB* = 2 and $\angle AOB = \frac{\pi}{3}$.

(a) Find **a** · **b**. ***(2 marks)***

(b) Find $\overrightarrow{OE}$ in terms of **a** and **b**. ***(5 marks)***
[Hint: Let *BE* : *EA* = *t* : (1– *t*).]

(c) *F* is a variable point on *OE*. A student says that $\overrightarrow{BA} \cdot \overrightarrow{BF}$ is always a constant.

Explain whether the student is correct or not.

If you agree with the student, find the value of that constant.

If you do not agree with the student, find two possible values of $\overrightarrow{BA} \cdot \overrightarrow{BF}$.

(5 marks)

HKCEE Questions (level 3)

Not for sale

HKCEE Questions (level 3)

3. ***[Add. Math 2000 Paper I Q.9]***

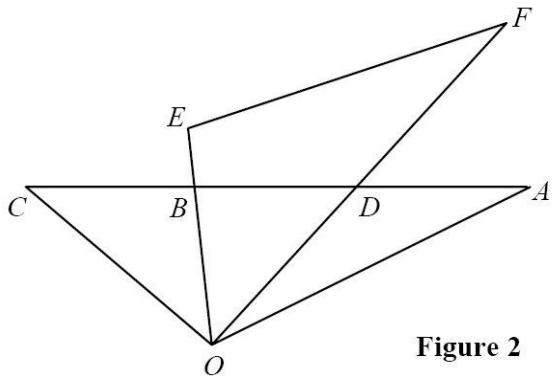


Figure 2

In Figure 2, OAC is a triangle. B and D are points on AC such that $AD = DB = BC$. F is a point on OD produced such that $OD = DF$. E is a point on OB produced such that $OE = k(OB)$, where $k > 1$. Let $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.

(a) (i) Express $\overrightarrow{OD}$ in terms of **a** and **b**.

(ii) Show that $\overrightarrow{OC} = -\frac{1}{2}\mathbf{a} + \frac{3}{2}\mathbf{b}$.

(iii) Express $\overrightarrow{EF}$ in terms of k, **a** and **b**. ***(5 marks)***

(b) It is given that $OA = 3$, $OB = 2$ and $AOB = 60°$.

(i) Find $\mathbf{a} \cdot \mathbf{b}$ and $\mathbf{b} \cdot \mathbf{b}$.

(ii) Suppose that $OEF = 90°$.

(1) Find the value of k.

(2) A student states that points C, E and F are collinear.
Explain whether the student is correct. ***(11 marks)***

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HKCEE Questions (level 3)

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HKCEE Questions (level 4)

4. ***[A.Math 1998 Paper I Q.9]***

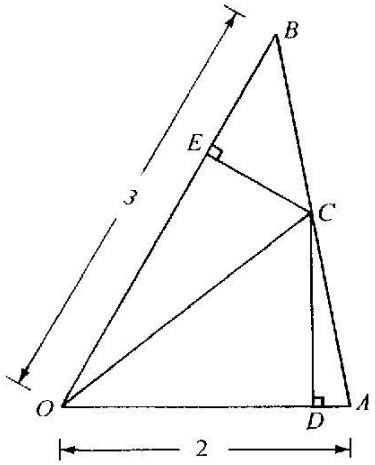


Figure 2

In Figure 2 , OAB is a triangle with $OA = 2$, $OB = 3$ and $\angle AOB = 60^\circ$. C is a point on AB such that $AC : CB = t : 1 - t$, where $0 < t < 1$. D and E are respectively the feet of perpendicular from C to OA and OB. Let $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.

(a) (i) Find $\mathbf{a} \cdot \mathbf{b}$.

(ii) Express $\overrightarrow{OC}$ in terms of t, $\mathbf{a}$ and $\mathbf{b}$.

(iii) Express $\mathbf{a} \cdot \overrightarrow{OC}$ and $\mathbf{b} \cdot \overrightarrow{OC}$ in terms of t. ***(7 marks)***

(b) (i) Using (a) (iii), show that $\mathbf{a} \cdot \overrightarrow{OD} = 4 - t$ and $\mathbf{b} \cdot \overrightarrow{OE} = 3 + 6t$.

(ii) If $\overrightarrow{OD} = k\mathbf{a}$ and $\overrightarrow{OE} = s\mathbf{b}$, express k and s in terms of t.

(6 marks)

(c) Find the value of t such that $\overrightarrow{DE}$ is parallel to $\overrightarrow{AB}$. ***(3 marks)***

HKCEE Questions (level 4)

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HKCEE Questions (level 4)

5. ***[Add. Math 2009 Q.14]***

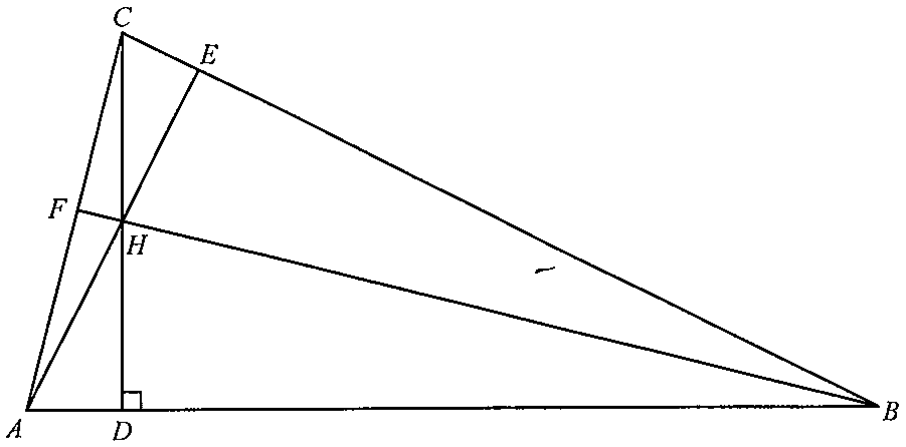


Figure 4

In Figure 4, *CD* is an altitude of ΔABC and *H* is the mid-point of *CD*. *AH* and *BH* are produced to meet *BC* and *AC* at *E* and *F* respectively.

Let **p**, λ **p** ($\lambda > 1$) and **q** be $\overrightarrow{AD}$, $\overrightarrow{AB}$ and $\overrightarrow{DH}$ respectively. Let $\frac{BE}{EC} = r$.

(a) Find $\overrightarrow{AH}$ in terms of **p** and **q**. ***(1 mark)***

(b) Express $\overrightarrow{AE}$ in terms of λ, r, **p** and **q**.

Hence show that r = λ. ***(4 marks)***

(c) It is given that | **p** | = 1, | **q** | = 2 and *H* is the orthocentre of ΔABC.

(i) Find $\overrightarrow{AE}$ in terms of **p** and **q**.

(ii) Find $\frac{AF}{FC}$. ***(7 marks)***

HKCEE Questions (level 4)

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HKCEE Questions (level 5)

6. ***[Add. Math 2008 Q.15]***

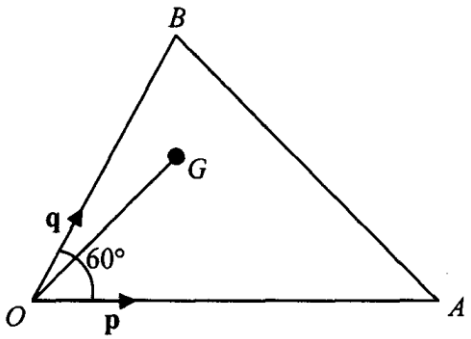


Figure 3

In Figure 3, **p** and **q** are unit vectors with angle between them 60°. Let $\overrightarrow{OA} = 4\mathbf{p}$, $\overrightarrow{OB} = 3\mathbf{q}$ and $\overrightarrow{OG} = \frac{2}{3}\mathbf{p} + \frac{5}{3}\mathbf{q}$.

(a) Find $\mathbf{p} \cdot \mathbf{q}$. ***(1 mark)***

(b) Show that $OG \perp AB$. Hence show that G is the orthocentre of ΔOAB. ***(5 marks)***

(c)

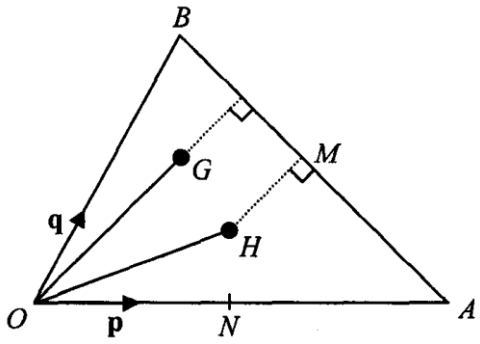


Figure 4

In Figure 4, H is the circumcentre of ΔOAB, M and N are the mid-points of AB and OA respectively. Let $HM : OG = t : 1$. By expressing $\overrightarrow{HM}$ and $\overrightarrow{HN}$ in terms of t, **p** and **q**, find $\overrightarrow{OH}$ in terms of **p** and **q**. ***(6 marks)***

HKCEE Questions (level 5)

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HKDSE Questions (Level 5)

7. ***[M2 2014 Q.11]***

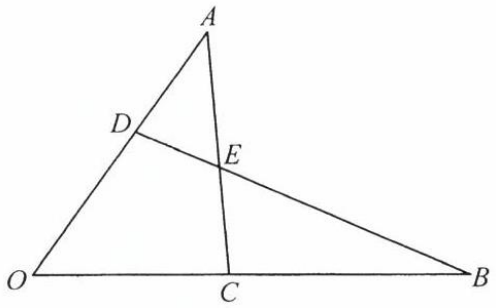


Figure 4

C and *D* are points on *OB* and *OA* such that $AD : DO = OC : CB = t : 1-t$, where $0 < t < 1$. *BD* and *AC* intersect at *E* such that *AE* : *EC* = *m* : 1 and *BE* : *ED* = *n* :1, where *m* and *n* are positive. Let $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.

(a) (i) By considering ΔOAC, express $\overrightarrow{OE}$ in terms of *m*, *t*, **a** and **b.**

(ii) By considering ΔOBD, express $\overrightarrow{OE}$ in terms of *n*, *t*, **a** and **b.**

(iii) Show that $m = \dfrac{t}{(1-t)^2}$ and $n = \dfrac{1-t}{t^2}$.

(iv) Chris claims that
"if *m* = *n*, then *E* is the centroid of ΔOAB".
Do you agree? Explain your answer. ***(9 marks)***

(b) It is given that *OA* = 1 and *OB* = 2. Francis claims that
"if *AC* is perpendicular to *OB*, then *BD* is always perpendicular to *OA*".
Do you agree? Explain your answer. ***(4 marks)***

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HKDSE Questions (Level 5)

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HKCEE Questions (level ∞)

8. ***[Add. Math 2006 Q.18]***

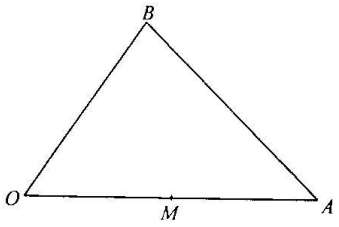


Figure 9

Figure 9 shows a triangle *OAB*. Let $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$ and *M* be the mid-point of *OA*.

(a)

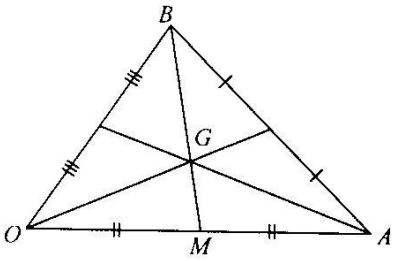


Figure 10

Let *G* be the centroid of ΔOAB (see Figure 10). It is given that *BG* : *GM* = 2 : Express $\overrightarrow{OG}$ in terms of ***a*** and ***b***. ***(1 mark)***

(b)

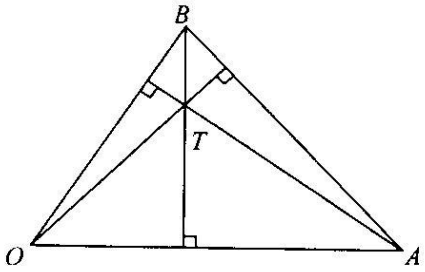


Figure 11

Let *T* be the orthocenter of ΔOAB (see Figure 11). Show that $\overrightarrow{OT} \cdot \mathbf{a} - \mathbf{b} \cdot \mathbf{a} = 0$ and write down the value of $\overrightarrow{OT} \cdot \mathbf{b} - \mathbf{a} \cdot \mathbf{b}$.

(3 marks)

HKCEE Questions (level ∞)

(c)

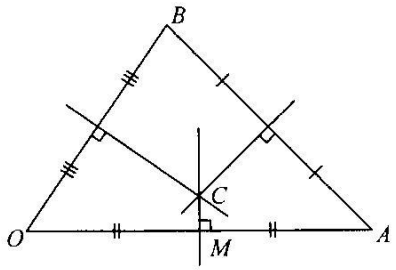


Figure 12

Let C be the circumcentre of ΔOAB (see Figure 12). Show that $2\overrightarrow{OC} \cdot \mathbf{a} = |\mathbf{a}|^2$ and find $\overrightarrow{OC} \cdot \mathbf{b}$ in terms of $|\mathbf{b}|$. ***(3 marks)***

(d) Consider the points G, T and C described in (a), (b) and (c) respectively.

(i) Using the above results, find the values of $(\overrightarrow{GT} - 2\overrightarrow{CG}) \cdot \mathbf{a}$ and $(\overrightarrow{GT} - 2\overrightarrow{CG}) \cdot \mathbf{b}$.

(ii) Show that G, T and C are collinear. ***(5 marks)***

Note : You may use the following property for vectors in the two-dimensional space.
If $\mathbf{w} \cdot \mathbf{u} = \mathbf{w} \cdot \mathbf{v} = 0$, where $\mathbf{u}$ and $\mathbf{v}$ are non-parallel, then $\mathbf{w} = 0$.

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More Questions in Past Papers

1. ***[Add. Math 2004 Q.6]***

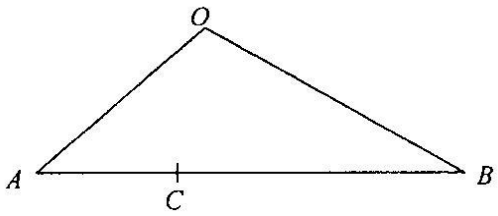


Figure 2

In Figure 2, *OAB* is a triangle. *C* is a point on *AB* such that *AC* : *CB* = 1 : 2. Let $\overrightarrow{OA}$ = **a**, $\overrightarrow{OB}$ = **b**.

(a) Express $\overrightarrow{OC}$ in terms of **a** and **b**.

(b) If |**a**| = 1, |**b**| = 2 and $\angle AOB = \frac{2\pi}{3}$, find $\left|\overrightarrow{OC}\right|$. ***(5 marks)***

2. ***[Add. Math 2005 Q.11]***

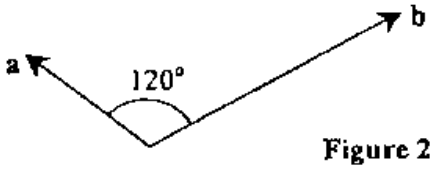


Figure 2

Figure 2 shows two vectors **a** and **b**, where |**a**| = 3, |**b**| = 5, and the angle between the two vectors is $120°$.

(a) Find **a** · **b**.

(b) Let **c** be a vector such that **a** + **b** + **c** = 0. Find |**c**|. ***(6 marks)***

3. ***[Add. Math 2010 Q.12]***

It is given that $\overrightarrow{OA} = \mathbf{i} + 2\mathbf{j}$, $|\overrightarrow{OB}| = 5\sqrt{2}$ and $\cos\angle AOB = \frac{3}{\sqrt{10}}$.

(a) Evaluate $\overrightarrow{OA} \bullet \overrightarrow{OB}$.

(b) Find $\overrightarrow{OB}$. ***(7 marks)***

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More Questions in Past Papers

4. ***[Add. Math 2002 Q.10]***

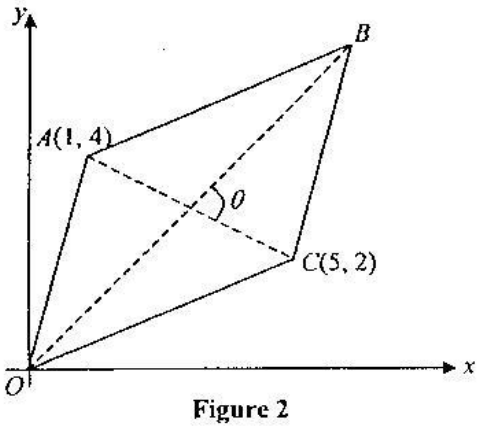


Figure 2

Figure 2 shows a parallelogram *OABC*. The position vectors of the points *A* and *C* are **i** + 4**j** and 5**i** + 2**j** respectively.

(a) Find $\overrightarrow{OB}$ and $\overrightarrow{AC}$.

(b) Let θ be the acute angle between *OB* and *AC*. Find θ correct to the nearest degree. ***(6 marks)***

5. ***[Add. Math 2003 Q.14]***

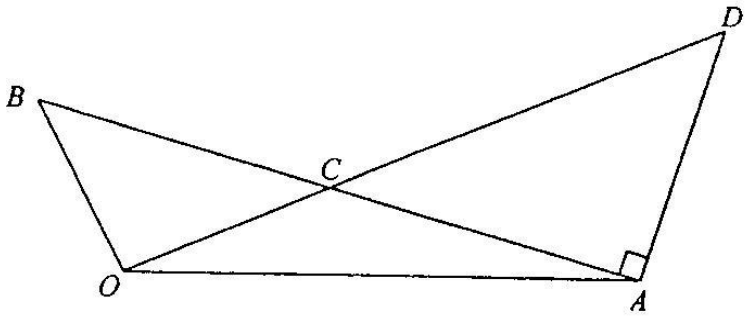


Figure 3

In Figure 3, *OAB* is a triangle such that *OA* = 3, *OB* = 1 and $\angle AOB = 120°$. *C* is a point on *AB* such that *AC* : *CB* = 3 : 2. *D* is a point on *OC* produced such that $\overrightarrow{OD} = k\overrightarrow{OC}$ and *AB* is perpendicular to *AD*. Let $\overrightarrow{OA}$ = **a** and $\overrightarrow{OB}$ = **b**.

(a) Find **a** · **b**. ***(2 marks)***

(b) Show that $\overrightarrow{AD} = (\frac{2k}{5} - 1)\mathbf{a} + \frac{3k}{5}\mathbf{b}$.

Hence find the value of *k*. ***(6 marks)***

(c) Determine whether the triangles *OCB* and *ACD* are similar. ***(4 marks)***

More Questions in Past Papers

6. ***[Add. Math 2007 Q.17]***

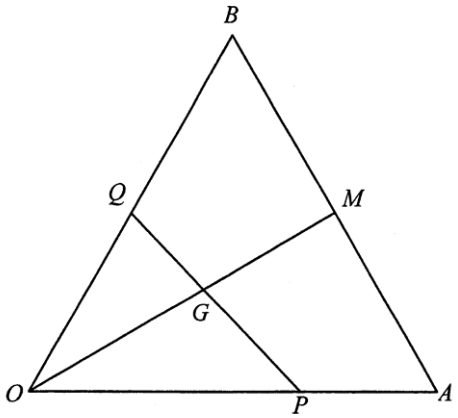


Figure 10

In Figure 10, *OAB* is an equilateral triangle with *OA* = 1. *M* is the mid-point of *AB* and *P* divides the line segment *OA* in the ratio 2 : 1 . Q is a point on *OB* such that *PQ* intersects *OM* at *G* and *PG* : *GQ* = 4 : 3. Let $\overrightarrow{OA}$ = **a** and $\overrightarrow{OB}$ = **b.**

(a) Find $\overrightarrow{OM}$ in terms of **a** and **b**. ***(1 mark)***

(b) Let *OQ* : *QB* = *k* : (1 – *k*)

(i) Find $\overrightarrow{OG}$ in terms of *k*, **a** and **b**.

(ii) Show that $\overrightarrow{PQ} = \frac{1}{2}\mathbf{b} - \frac{2}{3}\mathbf{a}$. ***(4 marks)***

(c) (i) Find $\mathbf{a} \cdot \mathbf{b}$ and hence find $\left|\overrightarrow{PQ}\right|$.

(ii) Find $\angle QGM$ correct to the nearest degree. ***(7 marks)***

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More Questions in Past Papers

7. ***[Add. Math 2010 Q.14]***

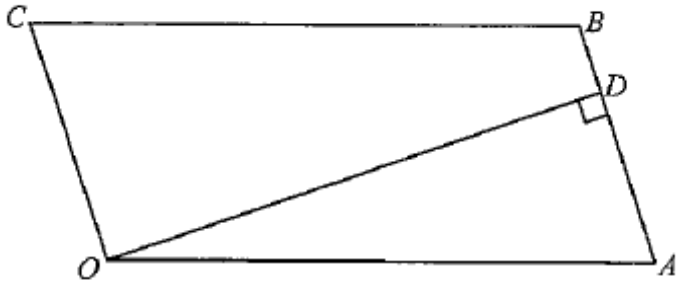


Figure 5

In Figure 5, *OABC* is a parallelogram with *OA* = 7, *OC* = 3 and $\angle AOC = \theta$ where $\cos\theta = \frac{-1}{3}$. *D* is a point on *AB* such that $OD \perp AB$ and $AD : DB = 1 : r$. Let $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OC} = \mathbf{c}$.

(a) By expressing $\overrightarrow{OD}$ in terms of **a**, **c** and *r*, find the value of *r*. ***(4 marks)***

(b) *E* is a point on *OD* produced such that *C*, *B* and *E* are collinear.

(i) Express $\overrightarrow{OE}$ in terms of **a** and **c**.

(ii) Are *A*, *O*, *C* and *E* concyclic? Explain your answer. ***(8 marks)***

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More Questions in Past Papers

8. ***[Add. Math 2004 Q.13]***

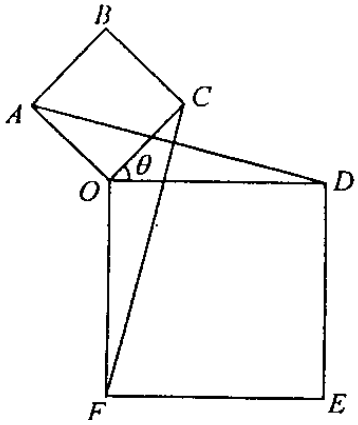


Figure 6

In Figure 6, *OABC* and *ODEF* are two squares such that *OA* = 1, *OF* = 2 and $\angle COD = \theta$, where $0° < \theta < 90°$. Let $\overrightarrow{OD} = 2\mathbf{i}$ and $\overrightarrow{OF} = -2\mathbf{j}$, where **i** and **j** are two perpendicular unit vectors.

(a) (i) Express $\overrightarrow{OC}$ and $\overrightarrow{OA}$ in terms of θ, **i** and **j**.

(ii) Show that $\overrightarrow{AD} = (2 + \sin\theta)\mathbf{i} - \cos\theta\mathbf{j}$. ***(4 marks)***

(b) Show that $\overrightarrow{AD}$ is always perpendicular to $\overrightarrow{FC}$. ***(4 marks)***

(c) Find the value(s) of θ such that points *B*, *C* and *E* are collinear. Give your answer(s) correct to the nearest degree. ***(4 marks)***

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